

Busy Bucket

Complete Project Documentation

Roles & Permissions · Panels · Data Models · Registration Approval Flows · Flutter
Director App

Version 3 — Detailed Edition

Cities covered: Mohali · Panchkula · Zirakpur · Ludhiana · Dehradun

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1. Introduction & Business Context

Busy Bucket ek professional home-cleaning service company hai jiska tagline hai "We clean, you relax". Company currently Chandigarh Tricity (Mohali, Panchkula, Zirakpur), Ludhiana, aur Dehradun mein operate karti hai, aur nearby areas (Rishikesh, Haridwar) ke liye bhi localized booking pages ban chuke hain.

Business do channels se customers tak pahunchti hai — public website (busybucket.in) jahan se customer directly ya WhatsApp ke through booking karta hai, aur ek internal operations system jo staff (Partners), unke supervisors (Admin/Manager), aur top-level leadership (Director) ko connect karta hai.

1.1 Digital System — Do Bade Hisse

- **Public-facing website** — marketing, service info, city-wise booking pages, WhatsApp integration.
- **Internal role-based panel system** — Director → Manager → Admin → Partner hierarchy, jisse bookings, staff, aur payments manage hote hain.

1.2 Is Document Ka Purpose

Yeh document poore Busy Bucket internal system ko explain karta hai — kaunsa panel kya karta hai, har role ke paas kya permissions hain, data kaise structure hoga, aur naya Director-panel Flutter app kis tarah kaam karta hai. Isi mein ek naya feature bhi cover kiya gaya hai: Director registration ka developer-approval flow.

2. System Architecture

High-level architecture teen layers mein socha gaya hai — Presentation (jo user dekhta hai), Application/Service (business logic), aur Data (storage). Neeche har layer detail mein hai.

2.1 Presentation Layer

Surface	Technology	Status
Public website	WordPress + custom HTML/CSS (Elementor/Divi scoped CSS)	Live
Role-selector landing page	Static HTML (hh.html)	Live
Admin panel	HTML page, stacked scroll layout	Live
Partner panel	HTML, mobile-first phone-frame layout	Live
Director panel	Flutter (Android/iOS/Web)	New — built
Manager panel	Flutter (planned, same pattern as Director)	Planned

2.2 Application / Service Layer

Flutter app mein service classes backend-agnostic rakhi gayi hain (e.g. AuthService, ManagerApprovalService, DashboardService) — abhi in-memory/mock data return karti hain, lekin unka interface aisa design kiya gaya hai ki Firestore ya kisi bhi REST API se seedha replace ho sake, bina UI code change kiye.

2.3 Data Layer

Abhi tak koi specific database finalize nahi hua hai. Recommended options:

- **Firebase / Firestore** — fastest to integrate with Flutter, real-time updates, built-in phone-OTP auth.
- **Custom REST API + PostgreSQL/MySQL** — agar existing WordPress/PHP backend ke saath integrate karna ho.

2.4 Authentication

Partner aur Director dono OTP-based phone login use karte hain. Manager aur Admin ke liye bhi same pattern follow hoga taaki poore system mein consistent login experience rahe.

3. Role Hierarchy & Permissions Matrix

System mein 4 roles hain, top se bottom tak strict hierarchy ke saath. Har level apne seedhe niche wale level ko manage/approve karta hai.

Level	Role	Approved By	Approves	Core Responsibilities
1	Director	Developer / Backend (manual)	New Managers	Overall business control, reports, final authority
2	Manager	Director (in-app)	New Admins	Area/city operations, Admin supervision
3	Admin	Manager / Director	New Partners	Booking assignment, partner management, payments
4	Partner	Admin	—	Field cleaning work, attendance, earnings

3.1 Permissions Matrix

Action	Director	Manager	Admin	Partner
View all-city reports	✓	Apne city tak	Apne city tak	✗
Approve Manager registration	✓	✗	✗	✗
Approve Admin registration	✓	✓	✗	✗
Approve Partner registration	✓	✓	✓	✗
Assign bookings	✗ (view only)	✗ (view only)	✓	✗ (receives only)
Mark attendance	✗	✗	✗	✓
View own earnings	✗	✗	✗	✓

4. Public Website (busybucket.in)

Design system: pine green / coral / cream color palette, Fraunces + Inter fonts, scoped CSS namespaces taaki Elementor/Divi builders ke andar embed karne par conflicts na ho.

4.1 Key Pages

- City-specific booking pages — Dehradun, Rishikesh, Haridwar, Chandigarh, Panchkula, Zirakpur, Ludhiana — har ek WhatsApp booking integration aur city-specific copy ke saath.
- About Us, Contact Us, Bathroom & Kitchen Cleaning service pages — Urban Company se inspired video showcase functionality.
- AC service promotional banners — responsive, container-query based.
- Homepage — kai baar redesign hua hai; WordPress theme mein convert kiya gaya (ACF + Customizer support), aur ek standalone HTML version bhi hai jo busybucket.in se live assets pull karta hai.

4.2 Technical Decisions (History)

- React/Next.js migration ko avoid kiya gaya — SEO equity protect karne ke liye WordPress par hi rehna decide hua.
- Spam attacks ke baad security hardening plan bana — WAF, 2FA, managed hosting.
- Branded HTML strategy document diya gaya tha — pricing on service cards, B2B vs B2C booking paths, city-detection prompts.

5. Admin Panel — Full Detail

Admin panel ek single, stacked scrolling page hai — teen major sections ek ke baad ek vertically stacked, jaisa hand-drawn sketch mein design kiya gaya tha.

5.1 Section — Abhi ki Booking

Live/ongoing bookings ke cards dikhata hai, real-time status ke saath (in-progress, on-the-way, completed). Client aur partner details ek hi combined card mein show hote hain (pehle alag-alag the, ab merge kar diya gaya).

5.2 Section — Booking Assign

Naye, unassigned bookings ki list. Har booking ke liye eligible partners ko notify kiya jaata hai aur ek 30-second accept/reject timer chalta hai — agar partner time par respond nahi karta to booking agle available partner ko route ho jaati hai. Admin yahin se manually bhi ek naya booking add kar sakta hai — form seedha isi queue mein push hota hai.

5.3 Section — Partners

Teen filterable tabs:

- **Fixed Partners** — permanent/verified staff.
- **Under Training** — naye partners jo training phase mein hain.
- **Partner Requests** — pending onboarding requests, jinhe Admin view-profile karke accept/reject kar sakta hai.

5.4 Partner Onboarding Screen

Admin-side "Partner Profile Creation" screen se naye partners onboard hote hain: service assign karna, naam + OTP se phone verify, Aadhaar/PAN document upload, edit permissions, aur active/deactivate status toggle.

6. Partner Panel — Full Detail

Mobile-first design, phone-frame layout ke saath — jaise ek actual phone screen ke andar poora UI render hota hai. OTP-based login.

6.1 Login Flow

- 1 Partner apna phone number enter karta hai.
- 2 OTP bhejaa jaata hai.
- 3 6-digit OTP verify karne par Dashboard khulta hai.

6.2 Screens

Screen	Purpose
Dashboard	Assigned jobs ka overview, notifications
Attendance	Daily check-in / check-out tracking
Payments	Earnings summary aur payment history
Profile	Partner details, uploaded documents

6.3 Navigation

Fixed bottom navigation bar poore panel mein consistent rehta hai — logo display aur brand asset background-removal issues already fix ho chuke hain.

7. Director Panel — Full Detail (Flutter App)

Director panel पूरी तरह Flutter mein bana diya gaya hai (busy_bucket_director_flutter.zip). Yeh section har screen, uske data flow, aur underlying code structure ko detail mein cover karta hai.

7.1 Login Screen

Phone number input → "OTP Bhejin" button → `AuthService.sendOtp()` call hota hai → OTP Screen par navigate.

7.2 OTP Screen

6-digit OTP input → `AuthService.verifyOtp()` → success par `Navigator.pushAndRemoveUntil` se Dashboard par redirect (poori login history clear ho jaati hai, back button se login screen par wapas nahi ja sakte).

7.3 Dashboard (Home Tab)

- Agar pending Manager approvals hain to coral-colored alert banner dikhta hai.
- 4 stat cards — Total Managers, Total Admins, Total Partners, Bookings Today.
- Pull-to-refresh se `DashboardService.getStats()` dobara call hota hai.

7.4 Manager Approvals Tab

Sab pending Manager registration requests ki list (naam, city, phone, submitted date). Har card par Accept/Reject button:

- **Accept** → `ManagerApprovalService.accept(id)` → account active, list se turant hat jaata hai, success snackbar dikhta hai.
- **Reject** → `ManagerApprovalService.reject(id)` → entry list se hat jaati hai, koi account create nahi hota.

7.5 Managers Tab

Active/approved Managers ki list — naam, city, phone number.

7.6 Profile Tab

Director details, Account Settings, Reports (placeholder), Logout.

7.7 File Structure

```
lib/main.dart
lib/theme/app_theme.dart
lib/models/director_model.dart
lib/models/manager_request_model.dart
lib/services/auth_service.dart
lib/services/manager_approval_service.dart
lib/services/dashboard_service.dart
lib/services/pdf_export_service.dart
lib/screens/login_screen.dart
lib/screens/otp_screen.dart
lib/screens/director_dashboard_screen.dart
lib/screens/manager_approvals_screen.dart
lib/screens/manager_list_screen.dart
lib/screens/director_profile_screen.dart
lib/widgets/stat_card.dart
lib/widgets/manager_request_card.dart
pubspec.yaml
```

8. Manager Panel — Planned Specification

Manager panel abhi build nahi hua hai, lekin Director panel jaisa hi pattern follow karega taaki code aur design consistent rahe.

Screen	Purpose
Login / OTP	Same flow jaisa Director panel mein hai
Dashboard	Apne area ke Admins/Partners/Bookings ka summary
Admin Approvals	Naye Admin registrations accept/reject (Director ke Manager Approvals jaisa)
Admins List	Apne area ke active Admins
Reports	City-level performance metrics

Code-reuse: ManagerApprovalService ka pattern copy karke AdminApprovalService banaya ja sakta hai — sirf model aur naming change hogi.

9. Registration & Approval Workflows

9.1 Director Registration (Developer-Approved)

- 1 Naya Director register page ke through form submit karta hai (naam, phone, email).
- 2 Request **Pending Approvals** queue mein backend par store hoti hai — sirf developer/super-admin ko visible.
- 3 Developer manually review karke Accept ya Reject karta hai.
- 4 **Accept** → entry pending queue se turant hat jaati hai, account active ho jaata hai, Director login kar sakta hai.
- 5 **Reject** → entry hat jaati hai, dobara registration karni padegi.

Yeh gatekeeping isliye zaroori hai kyunki Director ke paas sabse zyada control hota hai — koi bhi khud se account bana kar directly access nahi le sakta.

9.2 Manager Registration (Director-Approved)

- 1 Naya Manager registration form submit karta hai (naam, phone, email, city).
- 2 Request Director ke **Manager Approvals** tab mein dikhti hai.
- 3 Director Accept/Reject karta hai — Accept par account active, list se turant hat jaata hai.

9.3 Admin & Partner Registration (Planned)

Same pattern age bhi continue hoga — Admin registration Manager/Director approve karega, aur Partner registration Admin approve karega (jo already Admin panel ke "Partner Requests" tab mein exist karta hai).

9.4 Booking Lifecycle (Reference)

Registration approvals ke alawa, ek aur core workflow hai jo poore system ko jodta hai — booking ka lifecycle, jo Admin panel se shuru hoke Partner panel tak jaata hai:

Stage	Kahan Hota Hai	Detail
Booking Created	Admin Panel	Client details, service, address, date/time enter
Assign Queue	Admin Panel — Booking Assign	Eligible partners ko notify, 30-sec timer
Accepted	Partner Panel — Dashboard	Partner ne accept kiya, job unke dashboard par aa jaati hai
In Progress	Admin Panel — Abhi ki Booking	Live status card update hota hai
Completed	Partner Panel	Partner complete mark karta hai, payment queue mein jaata hai
Paid	Partner Panel — Payments	Earnings history mein reflect hota hai

9.5 Notifications & Communication

System ke andar timely communication bahut important hai, especially approval aur booking workflows ke liye. Neeche notification points list kiye gaye hain jo har role ko milne chahiye:

Event	Kisko Notification	Channel (suggested)
Naya Director registration submit hua	Developer/Super-admin	Email / Slack
Director registration accept/reject hui	Naya Director	SMS / App push
Naya Manager registration submit hua	Director	In-app banner + push
Manager registration accept/reject hui	Naya Manager	SMS / App push
Naya booking assign hua	Partner	App push, 30-sec timer ke saath
Booking complete hui	Admin	In-app update
Payment process hui	Partner	SMS / App push

Abhi tak notification system implement nahi hua — yeh Roadmap (Section 15) mein bhi cover kiya gaya hai.

9.6 Reports & Analytics (Planned)

Director aur Manager dashboards ke "Reports" section abhi placeholder hain. Future mein yeh metrics track karna proposed hai:

- Daily/weekly/monthly bookings count — city-wise breakdown.
- Partner performance — completed jobs, average rating, on-time acceptance rate.
- Revenue aur payments — total collected, pending payouts.
- Approval turnaround time — kitni jaldi Director/Manager naye registrations approve kar rahe hain.
- Partner attrition/retention — Under Training se Fixed Partner tak conversion rate.

9.7

Partner Onboarding & Training Flow (4 Rounds)

Naya Partner ka ID create hone ke turant baad, Partner Panel mein ek 4-round training/onboarding flow start ho jaata hai. Har round tabhi unlock hota hai jab pichhla round complete ho chuka ho — yeh sequence strict hai aur skip nahi kiya ja sakta.

ROUND 1

Work Videos + Company Rules

Partner ko service-wise work videos aur company ke rules/policies dikhaye jaate hain. Partner ko sab dekhna aur read karna hota hai — tabhi agla round unlock hota hai.

Field	Detail
Content shown	Work / service videos (per assigned service) + company rules & policies
Partner action	Sab videos dekhna + rules read karna (mandatory, skip nahi ho sakta)
Unlocks	Round 2, jab sab content viewed/read mark ho jaaye

ROUND 2

Admin Meeting Details

Partner ko batata hai ki wo kis Admin se milega — meeting ki date, time, aur location ke saath us Admin ka naam bhi show hota hai.

Field	Detail
Fields shown	Meeting date, meeting time, location, aur assigned Admin ka naam
Source	Admin/backend se assign kiya gaya meeting slot
Unlocks	Round 3, jab partner meeting detail confirm/acknowledge kare

ROUND 3

Edit Details Form

Partner apni mandatory details fill/edit karta hai aur required documents upload karta hai (is form ka structure existing Partner Onboarding request-form jaisa hai — Aadhaar Card, Profile Photo, aur Machinery Photo upload set).

Field	Detail
Form type	Edit-details form — mandatory fields + document upload
Documents	Aadhaar Card, Profile Photo, Machinery Photo
Unlocks	Round 4, jab form submit ho jaaye

ROUND 4

Training Duration (Admin-set)

Rounds 1–3 complete hone ke baad hi Round 4 visible hota hai. Yahan partner ki training duration (kitne din/ghante) dikhai jaati hai — yeh duration Admin ke haath mein hoti hai: Admin isse set karta hai aur zaroorat padne par dobara add/adjust bhi kar sakta hai.

Field	Detail
Visible after	Round 1, 2 aur 3 teeno complete hone ke baad
Content shown	Training duration (day/hours) — Admin dwara set
Admin control	Admin duration ko set kar sakta hai aur baad mein dubara add/adjust kar sakta hai
Completion trigger	Admin-set training duration पूरी होने पर flow complete

Post-Training — Dashboard Activation

Round 4 ki training duration complete hone ke baad, Partner ka dashboard panel properly activate ho jaata hai. Is point ke baad partner apna poora kaam is dashboard se hi manage karta hai — jobs, attendance, aur **Payouts** section bhi ismein neeche (down area) visible ho jaata hai.

Note: Yeh flow existing 4-stage sketch (training videos → company policies → package chart → mandatory fields) ka updated/finalized version hai — location/contact step mein date aur time add kiya gaya hai, aur request-form step ke single equipment upload ko Aadhaar Card / Profile Photo / Machinery Photo document-upload set se replace kiya gaya hai.

10. Data Models — Full Field Reference

Model	Fields
DirectorModel	id, name, phone, email, activeSince
ManagerRequest	id, name, phone, email, city, submittedAt, status (pending/accepted/rejected)
DashboardStats	totalManagers, totalAdmins, totalPartners, activeBookingsToday, pendingManagerApprovals
DirectorRegistrationRequest	id, name, phone, email, submittedAt, status
AdminRequest (planned)	id, name, phone, email, city, submittedAt, status
PartnerModel (existing)	id, name, phone, servicesAssigned, aadhaarDoc, panDoc, status (fixed/training/requested)
BookingModel (existing)	id, clientName, address, service, assignedPartnerId, status, timerSeconds

11. Flutter App — Technical Architecture

11.1 Theming

app_theme.dart mein poore website jaisa hi pine green / coral palette define kiya gaya hai, taaki Director app ka look poore brand ke saath consistent rahe (AppColors.pineDark, AppColors.pine, AppColors.coral).

11.2 State Management

Abhi simple StatefulWidget + setState use kiya gaya hai — chhote app ke liye kaafi hai. Jab app bada ho (Manager, Admin, Partner sab Flutter mein aa jaye), Provider ya Riverpod jaisa state management add karna recommended hoga.

11.3 Packages Used

Package	Purpose
pdf	Documentation/report PDF generate karna app ke andar se
printing	Generated PDF preview/share/print karna
cupertino_icons	iOS-style icons

11.4 Design Decisions — Why This Structure

Code ko models / services / screens / widgets mein separate rakha gaya hai (clean layering) taaki:

- UI (screens/widgets) aur business logic (services) alag rahein — testing aasan ho.
- Backend badalna ho (mock se Firestore) to sirf services/ folder touch karna pade, UI wahi rahe.
- Manager, Admin, Partner panels banate waqt models aur pattern reuse ho sake, duplicate code na likhna pade.

11.5 Backend Integration Point

Har service class (AuthService, ManagerApprovalService, DashboardService) mein // TODO comments diye gaye hain jahan real backend (Firestore/REST) plug karna hai — UI code ko touch karne ki zaroorat nahi padegi.

12. Security & Access Control Considerations

- OTP-based login se password-related risks kam hote hain, lekin rate-limiting zaroori hai (OTP spam/abuse rokne ke liye).
- Role-based access control (RBAC) backend par bhi enforce hona chahiye — sirf app-level UI restriction kaafi nahi hai.
- Director registration approval developer ke paas hone se ek extra manual security gate milta hai.
- Aadhaar/PAN documents jaise sensitive data ke liye encrypted storage aur restricted access zaroori hai.
- Pehle spam attacks ho chuke hain website par — isi wajah se WAF, 2FA, aur managed hosting recommend kiya gaya tha; wahi principles internal panels par bhi apply hone chahiye.

12.1 Common Threat Scenarios & Mitigations

Scenario	Mitigation
Koi fake Director account banane ki koshish kare	Developer-side manual approval gate
Koi fake Manager account banane ki koshish kare	Director-side in-app approval
OTP brute-force / spam	Rate limiting + OTP expiry (e.g. 5 min)
Sensitive documents (Aadhaar/PAN) leak	Encrypted storage, access-logged downloads
Unauthorized API access	Backend-side RBAC checks, not just app-level UI hiding

13. Testing Plan

- **Unit tests** — service classes (AuthService, ManagerApprovalService) ke liye, mock data ke saath.
- **Widget tests** — Login, OTP, aur Manager Approvals screens ke UI states (loading, error, empty list) ke liye.
- **Manual QA** — real device par OTP flow, timer-based booking assignment, aur accept/reject actions test karna.
- **Regression check** — naya panel (Manager) add hone par Director panel ke shared components (theme, cards) na toote, yeh verify karna.

14. Deployment & Rollout Plan

- 1 Backend finalize karna (Firestore ya REST API) aur sab service classes ko connect karna.
- 2 Director app internal testing (developer + Director khud) ke liye distribute karna.
- 3 Real OTP provider integrate karna (abhi mock hai).
- 4 Play Store / App Store internal track par upload karna, ya direct APK distribute karna shuru mein.
- 5 Manager panel build hone ke baad wahi rollout process repeat karna.

15. Next Steps / Roadmap

- Manager panel Flutter app banana (Admin approvals ke saath) — isi codebase pattern par.
- Admin panel ko HTML se Flutter mein migrate karna.
- Partner panel ko HTML se Flutter mein migrate karna.
- Real backend connect karna — abhi sab services in-memory/mock hain.

- Push notifications add karna (naya booking, naya approval request, etc.).

16. Appendix — Glossary

Term	Meaning
Partner	Field staff member (cleaner) jo bookings complete karta hai
Admin	Bookings assign karne wala aur partners manage karne wala role
Manager	City/area-level supervisor, Admins ke upar
Director	Top-level role, poore business ka control
Pending Approvals Queue	Naye registration requests ki list jo approve/reject hone ka wait kar rahi hai
OTP	One-Time Password — phone-based login verification
RBAC	Role-Based Access Control

16.1 Frequently Asked Questions

Q: Director khud apni registration approve kyun nahi kar sakta?

Kyunki Director ke paas sabse zyada system control hota hai, is level par ek manual, external (developer-side) check rakha gaya hai taaki koi unauthorized account na ban sake.

Q: Agar Manager approval reject ho jaaye to kya user dobara try kar sakta hai?

Haan — reject hone par entry queue se hat jaati hai aur user dobara fresh registration submit kar sakta hai.

Q: Kya Admin aur Partner panels bhi Flutter mein aayenge?

Haan, Section 15 (Next Steps) mein yeh already roadmap mein hai — same design pattern follow karte hue.

Q: Backend abhi kaunsa use ho raha hai?

Abhi koi fixed backend nahi hai — Flutter services mock/in-memory data use karte hain, jise Firestore ya REST API se replace kiya ja sakta hai.

16.2 Document Version History

Version	Change
v1	Initial project overview PDF — Admin, Partner panels, basic registration flow
v2	Director Panel Flutter app add kiya, expanded detail
v3	Full detailed edition — architecture, permissions matrix, booking lifecycle, notifications, reports, testing, deployment